

DATA301/DATA471 - Assignment IV

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1 Exercise 1 - Estimators for the Exponential rate lambda

$X_1, \dots, X_n$ are independent and identically distributed as $\text{Exponential}(\lambda)$, where λ is greater than 0 and equals the reciprocal of the mean. I compare the maximum likelihood estimator, the inter-quartile-range (IQR) analogy estimator, and a bootstrap bias-corrected IQR estimator.

```
# Maximum likelihood: lambda_ML = 1 / mean(X)
lambda_ml <- function(x) 1 / mean(x)

# Fast type-7 quantile (identical to quantile(x, p, type = 7)), for speed.
q7 <- function(xs, p) {
  # xs already sorted
  n <- length(xs); h <- 1 + (n - 1) * p; lo <- floor(h); fr <- h - lo
  if (lo >= n) xs[n] else xs[lo] + fr * (xs[lo + 1] - xs[lo])
}

# IQR estimator: lambda_IQR = ln(3) / (Q3 - Q1), since Q3 - Q1 = ln(3)/lambda.
lambda_iqr <- function(x) { xs <- sort(x); log(3) / (q7(xs, 0.75) - q7(xs, 0.25)) }
```

1.1 Which definition of the sample quantiles? [10%]

Wicklin (2017) reviews nine definitions of the sample quantile: three discrete (which round to order statistics) and six continuous (which interpolate linearly between them). They differ mainly for small samples (n of 10 or fewer) and in the tails, and become indistinguishable by about $n = 100$. Defaults vary by software (SAS uses type 2; R uses type 7). Wicklin does not recommend a single type, but points to Hyndman and Fan (1996) for anyone who wants an unbiased or distribution-free choice. Hyndman and Fan recommend type 8, which is approximately median-unbiased for any distribution.

I use type 7, which is R's default (NumPy's "linear" method is the same). For continuous Exponential data with no ties the difference between the types is of order $1/n$ and cannot affect the comparison of estimators below, so the default is fine.

1.2 A bootstrap bias-corrected IQR estimator [20%]

The IQR estimator is a nonlinear function of two quantiles, so it is biased in finite samples. Write θ_{hat} for $\text{lambda_IQR}(X)$. The bootstrap (Efron, 1979) estimates the bias as $(\text{mean of } \theta_{\text{star}}) - \theta_{\text{hat}}$, where $\theta_{\text{star}_1}, \dots, \theta_{\text{star}_B}$ are the estimator recomputed on resamples of the data. The first-order bias-corrected estimator subtracts that estimate of the bias:

$$\text{lambda_BC} = \theta_{\text{hat}} - (\text{mean}(\theta_{\text{star}}) - \theta_{\text{hat}}) = 2 * \theta_{\text{hat}} - \text{mean}(\theta_{\text{star}}),$$

with $B = 200$ bootstrap resamples.

```
lambda_iqr_bc <- function(x, B = 200) {  
  theta_hat <- lambda_iqr(x)  
  n <- length(x)  
  theta_star <- replicate(B, lambda_iqr(x[sample.int(n, n, replace = TRUE)]))  
  2 * theta_hat - mean(theta_star)  
}
```

1.3 Monte Carlo comparison [20%]

Over λ in $\{0.1, 1, 10\}$ and n in $\{5, 30, 100\}$, with R replications per cell, I draw a fresh sample each replication and record the mean squared error (the sole metric). The MSE estimate for a cell is the average over the R replications of $(\lambda_{\text{hat}} - \lambda)^2$. I also record its Monte Carlo standard error.

```
lambdas <- c(0.1, 1, 10); ns <- c(5, 30, 100)  
R_MC <- 1000  
  
run_cell <- function(lambda, n, R = R_MC) {  
  est <- matrix(NA_real_, R, 3, dimnames = list(NULL, c("ML", "IQR", "IQR_BC")))  
  for (r in seq_len(R)) {  
    x <- rexp(n, rate = lambda)  
    est[r, ] <- c(lambda_ml(x), lambda_iqr(x), lambda_iqr_bc(x))  
  }  
  sq <- (est - lambda)^2  
  data.frame(lambda, n, estimator = colnames(est),  
             mse = colMeans(sq), mc_se = apply(sq, 2, sd) / sqrt(R),  
             row.names = NULL)  
}
```

```

results <- do.call(rbind, Map(run_cell,
                             expand.grid(lambda = lambdas, n = ns)$lambda,
                             expand.grid(lambda = lambdas, n = ns)$n))

mse_wide <- reshape(results[, c("lambda", "n", "estimator", "mse")],
                    idvar = c("lambda", "n"), timevar = "estimator", direction = "wide")
names(mse_wide) <- sub("^mse\\.", "", names(mse_wide))
knitr::kable(mse_wide, digits = 4, caption = "Estimated MSE by estimator for each (lambda, n).")

```

Table 1: Estimated MSE by estimator for each (lambda, n).

	lambda	n	ML	IQR	IQR_BC
1	0.1	5	0.0040	0.2579	Inf
4	1.0	5	0.7479	34.4390	Inf
7	10.0	5	68.1230	1502.3030	Inf
10	0.1	30	0.0004	0.0011	0.0013
13	1.0	30	0.0406	0.1370	0.1445
16	10.0	30	3.7962	9.9693	10.8417
19	0.1	100	0.0001	0.0003	0.0003
22	1.0	100	0.0103	0.0252	0.0295
25	10.0	100	1.1495	2.7896	3.1701

```

ggplot(results[is.finite(results$mse), ],
        aes(factor(n), mse, colour = estimator, group = estimator)) +
  geom_line() + geom_point() +
  facet_wrap(~ lambda, scales = "free_y", labeller = label_both) +
  scale_y_log10() +
  labs(x = "Sample size n", y = "MSE (log scale)", colour = "Estimator") +
  theme_minimal()

```

Replications. Following Robert and Casella (2009, section 3.2), a Monte Carlo average converges at a rate of order 1 over the square root of R, and its error is judged from the estimated standard error: the estimate plus or minus two standard errors gives an approximate 95 percent band by the Central Limit Theorem (section 4.2 makes the same assessment by monitoring the variability of the runs). Applying that criterion, the reported `mc_se` is under 10 percent of its MSE for every cell with `n` of 30 or more, which is far smaller than the gaps between the estimators, so `R = 1000` is enough there. A larger `R` cannot help at `n = 5`, where the estimator is undefined rather than merely noisy. This value of 1000 also matches the working default (`Nsim = 103`) used throughout Robert and Casella.

Results. ML has the lowest MSE in every cell, which is expected because it is efficient when the model is correct. The IQR estimator is about 2.4 to 3.4 times worse than ML at `n` of 30 or more, since it uses only the quartiles. The bias correction slightly increases the MSE (it removes bias but adds more variance), and this holds in every finite cell. At `n = 5` the IQR estimators fail: IQR is extremely variable, and IQR-BC is infinite, because bootstrap resamples of five values often give Q3 equal to Q1 (a zero denominator). The MSE scales like lambda squared across the columns and falls like 1 over `n` down the rows, as expected.

2 Exercise 2 - Time-series analysis of ausbeer [50%]

This is approached as exploratory data analysis (per the course announcement): the plots below extract the information in the data, and no model is fitted. The `ausbeer` series is quarterly Australian beer production in megalitres; I use the span from 1992 onward.

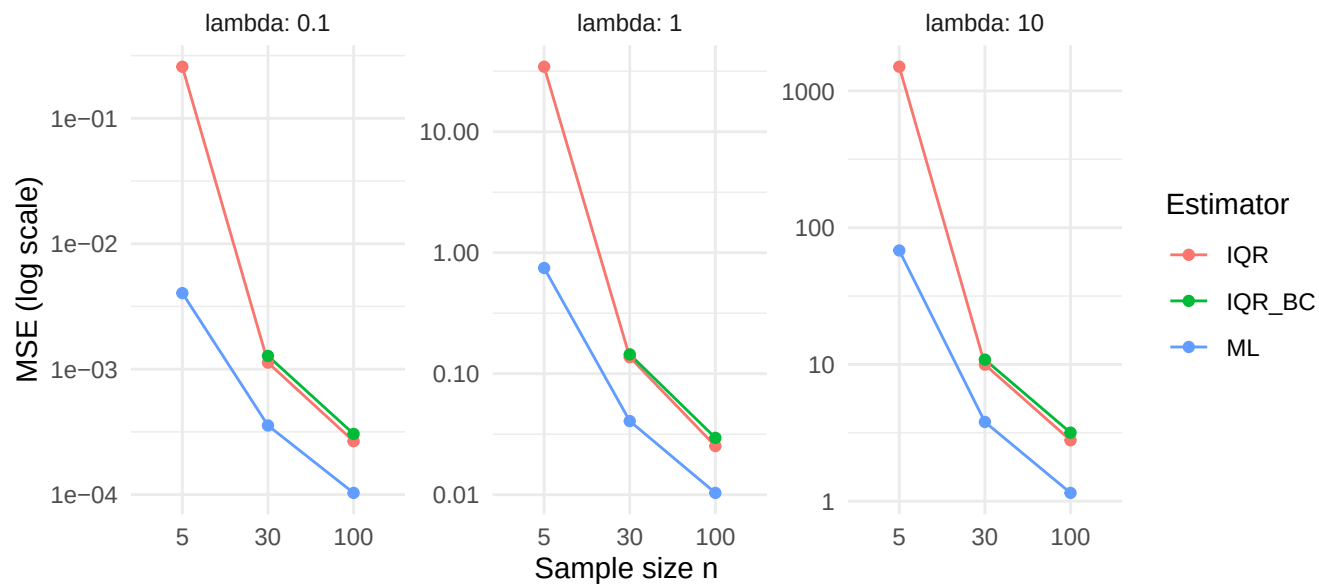


Figure 1: MSE against sample size, faceted by lambda (log scale). Non-finite points (IQR-BC at $n=5$) omitted.

```
beer <- window(ausbeer, start = 1992)
```

2.1 Frequency

```
frequency(beer)
```

```
## [1] 4
```

The frequency is 4 (quarterly); the plot also repeats every four observations.

2.2 Behaviour, trend, seasonality, variance, outliers

```
autoplot(beer) + labs(x = "Year", y = "Megalitres") + theme_minimal()
```

- Trend: no strong trend, just a slight downward drift of about 1.3 megalitres per year.
- Seasonality: clear and periodic (a fixed four-quarter cycle, not merely cyclic). The quarter means are Q1 429, Q2 394, Q3 411, Q4 502, so Q4 is highest and Q2 is lowest.
- Variance: the seasonal amplitude is stable over time, so no transformation is needed.
- Unusual points: 2010 Q2 (374 megalitres) is the lowest and 1992 Q4 (532 megalitres) the highest; both are consistent with the seasonal pattern rather than being data errors.

2.3 Seasonal plot

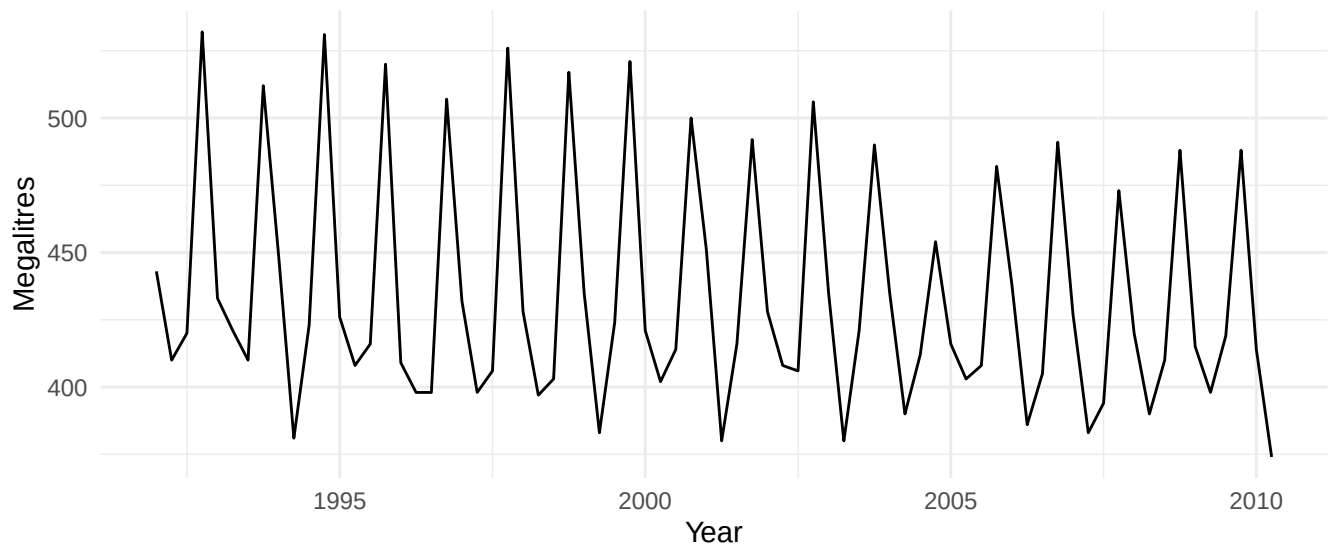


Figure 2: Quarterly Australian beer production, 1992 onward.

```
ggseasonplot(beer, year.labels = TRUE) + labs(y = "Megalitres") + theme_minimal()
```

Every year follows the same profile (a peak in Q4 and a trough in Q2) with roughly parallel lines, so the seasonal shape is stable and the years differ mainly in their overall level.

2.4 Seasonal subseries plot

```
ggsubseriesplot(beer) + labs(y = "Megalitres") + theme_minimal()
```

Q4 is the highest quarter and Q2 the lowest (the blue quarter means). Each quarter drifts down slightly over time, but the ranking never changes, so the seasonal pattern is stable.

2.5 Detrend or transform before modelling?

No. The variance is stable (so no transformation) and the trend is weak; the only substantial departure from stationarity is the seasonality, so if anything were done before modelling it would be a single seasonal difference. This is a visual argument only, with no model fitted.

3 References

- Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1), 1-26.
- Hyndman, R. J., and Athanasopoulos, G. (2018). *Forecasting: Principles and Practice*. OTexts. <https://OTexts.com/fpp2>

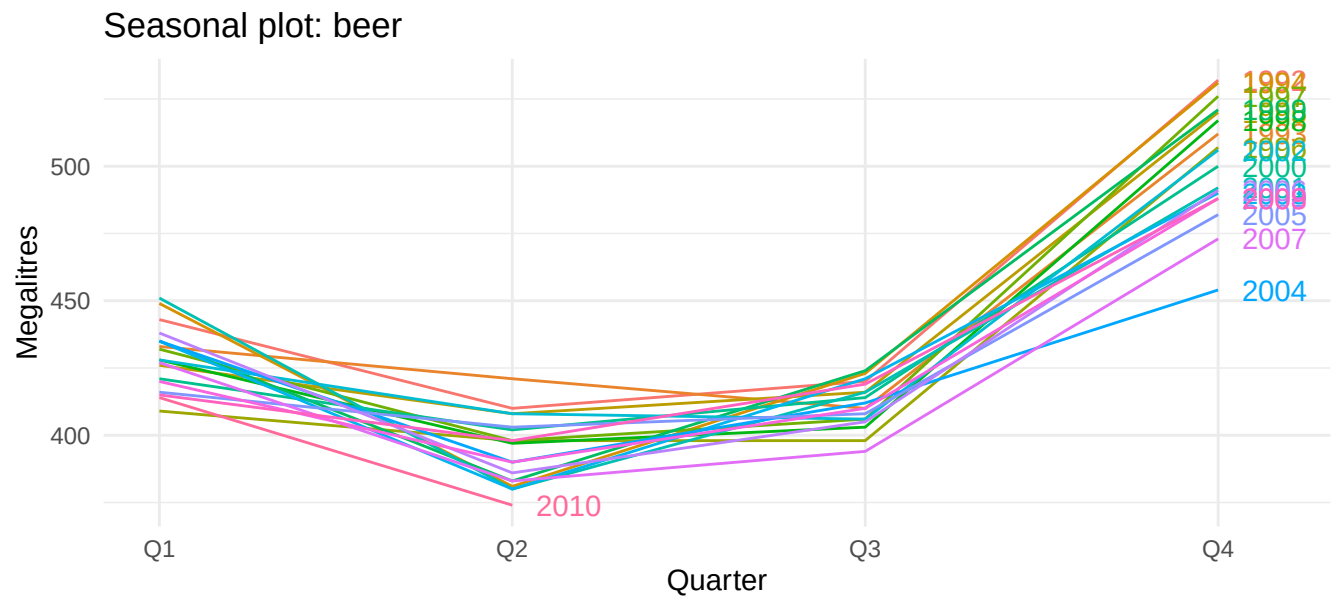


Figure 3: Seasonal plot.

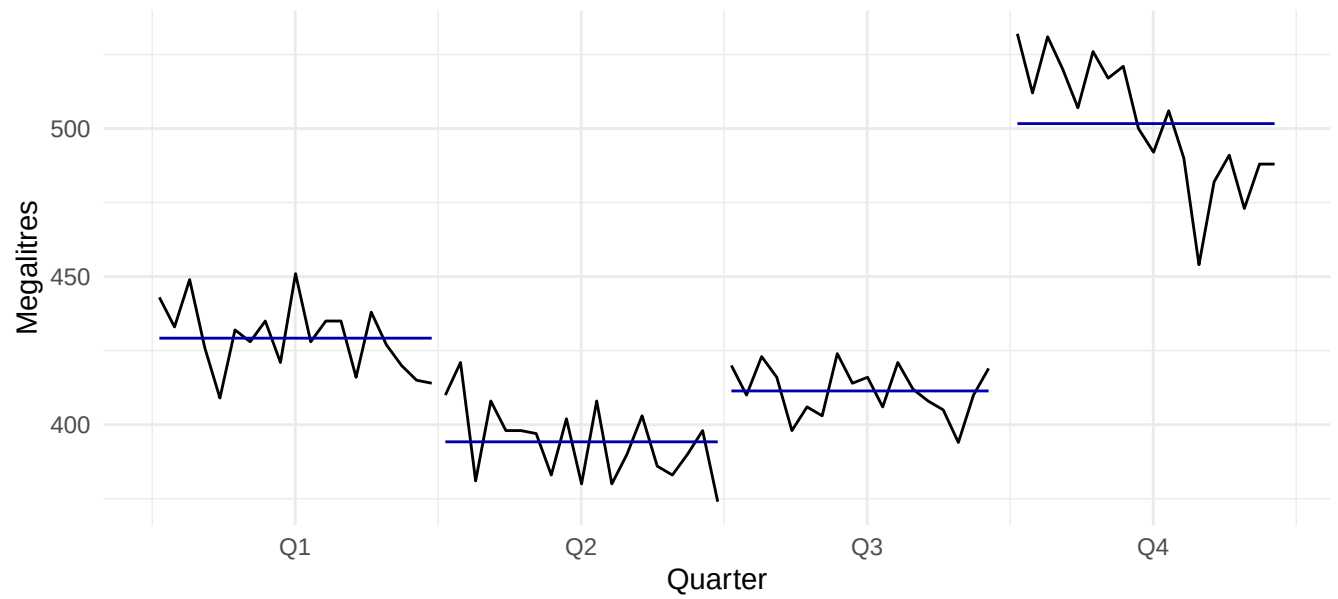


Figure 4: Seasonal subseries plot.

- Hyndman, R. J., and Fan, Y. (1996). Sample quantiles in statistical packages. *The American Statistician*, 50(4), 361-365.
- Robert, C. P., and Casella, G. (2009). *Introducing Monte Carlo Methods with R*. Springer.
- Wicklin, R. (2017). Sample quantiles: A comparison of 9 definitions. SAS Blogs. <https://blogs.sas.com/content/iml/2017/05/24/definitions-sample-quantiles.html>